RANDOM VARIABLES **28/04/26**

The below definitions are not ones you need to learn but will give you a sense of what the terms in italics mean as you will hear me use them a lot from now on. The idea of expectation is not actually on your syllabus but is a useful way of thinking about random variables and hence is included here.

- A *random variable* is a function mapping events in some sample space to some (usually numerical quantity). For example, we can define X to be the random variable denoting the sum of the two outcomes when two fair dice are rolled. The random variable is then effectively converting each event into a number so if two fours are rolled, the random variable outputs an 8.
- The *probability distribution* of a random variable is the set of all possible (mutually exclusive) outcomes of the random variable with their associated probabilities.
- The *expected value* of a random variable is the long run average value. For example, the expected outcome when rolling a fair die is 3.5 as that is the likely long run average roll.

  The formula for the expected value of a random variable X is given by:

$$\mathbb{E}(X) = \sum_r r \times \mathbb{P}(X = r)$$

where the sum is taken over all possible values of X.

PROBLEMS:

1. A random variable X has a probability distribution given by

$$\mathbb{P}(X = r) = \begin{cases} kr^2 & \text{if } r = 1,2,3,4; \\ 0 & \text{otherwise.} \end{cases}$$

   a. Calculate the value of k.
   b. Find $\mathbb{E}(X)$.

2. In a carnival game of 'skill' you need throw a hoop over bottles with different shaped tops. If you successfully get the hoop over the bottle then you win money, depending on how challenging the bottle is to hit. If you hit a 'normal' bottle then you win £5. If you hit an 'epic' bottle you win £10 and if you hit the 'legendary' bottle you win £20. Kiara likes to throw all the hoops at the same time, in a somewhat scattered way, and she believes that her probability that a single hoop will hit the easy bottle is 0.1, the epic bottle 0.05 and the legendary bottle 0.01. It costs £1 for three hoops.
   a. Explain how much Kiara thinks she is going to win, on average, per pound she spends on the game.
   b. Suggest why, even if Kiara plays this game many times, her actual average winnings might not be close to the answer you gave in Part a.

3. Three dice are thrown. The value of M is the largest score shown.
   a. What is the most likely value of M?
   b. What is the expected value of M?

4. Four fair dice are rolled. S is the number of sixes showing.
   a. Derive the probability distribution of S and find the expected value of S.
   b. You win £30 for each six, and the game costs £25 a go. What is your expected *profit*?

5. You have five coins in your pocket: two 10p coins, two 20p coins and a 50p coin.
   a. You take out one coin. Find the expected value of the coin
   b. You take out two coins. Find the expected total value of the two coins.
   c. You take out three coins. Find the expected total value of the three coins.

6. A fair dice is rolled 15 times. X models the number of times that a score of one is rolled.
    a. Write down the probability distribution of X.
    b. What is the expected value of X.
    c. Find the probability that fewer ones than expected are rolled.

7. You draw five cards from a pack of sixteen, consisting of the A, K, Q, J of each suit. You win an amount, in pounds, equal to the number of aces you draw.

    Derive the probability distribution of your winnings in the two cases:
    a. you draw five cards simultaneously;
    b. after you draw each card, it is returned to the pack and the pack thoroughly shuffled.

    Find the expectation of your winnings in each case.